



D.Y. Patil Academic Education Excellence Federation's

Dr. D. Y. Patil Technical Campus

Polytechnic | Engineering | MCA

(Approved by AICTE, Recognized by DTE, Govt. of Maharashtra, Affiliated to Savitribai Phule Pune University, Pune)

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DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

TOPIC: Computer Networks

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No.	Topic	Main Areas Covered
01	Introduction to Computer Networks	Meaning, importance and purpose
02	Network Types and Topologies	LAN, MAN, WAN, star, mesh and bus topologies
03	OSI Reference Model	7 layers, architecture and data transmission flow
04	TCP/IP Protocol Architecture	4 layers, comparison with OSI model
05	Transmission Media	Guided (twisted pair, coaxial, fiber) and unguided
06	Data Link Layer & MAC Sublayer	Framing, error detection, flow control and CSMA/CD
07	Network Layer & IP Addressing	IPv4, IPv6, classful addressing, subnetting and CIDR
08	Routing Protocols	Distance vector, link state, RIP, OSPF and BGP
09	Transport Layer Protocols	TCP vs UDP, 3-way handshake and flow control
10	Application Layer Services	DNS, HTTP/HTTPS, FTP, SMTP and DHCP services
11	Network Security Fundamentals	Firewalls, cryptography, SSL/TLS and VPNs
12	Network Troubleshooting & Devices	Switches, routers, ping, traceroute and netstat
13	Conclusion	Key points and importance of computer networks



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INTRODUCTION TO COMPUTER NETWORKS

What is a Computer Network?

A computer network is an interconnected collection of autonomous computing devices, communication links, and networking hardware that exchange data and share digital resources. It encompasses both the physical transmission infrastructure and the communication protocols that enable seamless interaction between heterogeneous systems.

A robust computer network enables organizations and individuals to transfer data rapidly, access centralized applications, collaborate across geographical distances, and distribute processing workloads with high reliability and fault tolerance.

Importance of Computer Networks

- Facilitates seamless hardware, software, and data resource sharing.
- Enables instantaneous communication across distributed environments.
- Supports centralized data management, backup, and security governance.
- Lowers overall operational infrastructure costs through virtualization.
- Serves as the vital backbone for modern cloud computing and web services.

Example

Basic: Two standalone computers exchanging files using an external flash drive.

Better: Multiple workstations connected via high-speed gigabit switches sharing a centralized storage area network (SAN) and automated cloud backup.

The networked approach provides instantaneous synchronization, strict access control, and eliminates physical media bottlenecks.



TYPES OF NETWORKS AND TOPOLOGIES

1. Local Area Network (LAN)

A Local Area Network spans a limited geographical boundary such as an office, laboratory, or college campus. It provides high data-transfer rates and low propagation delay with private ownership of communication links.

Examples: Campus Wi-Fi network, departmental computer laboratory, corporate office Ethernet.

2. Wide Area Network (WAN)

A Wide Area Network covers extensive geographical spans such as countries or continents. WANs typically interconnect multiple LANs using public telecommunications infrastructure, leased fiber-optic links, or satellite relays.

Example: The global Internet is the largest public WAN, interconnecting millions of enterprise, governmental, and private networks.

3. Network Topologies

Network topology defines the physical or logical arrangement of computing nodes and interconnected transmission channels.

Topology	Key Characteristics & Practical Application
Star	All nodes connect to a central switch; easy to troubleshoot; standard for enterprise LANs.
Mesh	Dedicated point-to-point links between all nodes; highest fault tolerance; used in core WAN backbones.
Bus	Single shared coaxial backbone cable; low cabling cost; single break halts entire network.
Ring	Nodes form a closed circular loop; deterministic token passing; used in specialized industrial networks.

Why the Types Matter

- LANs optimize localized high-speed bandwidth at minimal operational costs.
- WANs enable global connectivity and distributed resource accessibility.
- Topology selection directly dictates fault tolerance, scalability, and maintenance costs.



OSI REFERENCE MODEL

The Open Systems Interconnection (OSI) model, developed by the ISO, provides a conceptual 7-layer architectural framework for standardizing telecommunication and computing protocols across diverse physical hardware.

Upper Layers (Application & Host Oriented)

The upper layers manage application-level semantics, user sessions, data syntax, and presentation formats.

Layer	Primary Function	Protocols / Data Unit
7. Application	Direct interface to user software and network services	HTTP, DNS, FTP (Data)
6. Presentation	Data encryption, compression, and format translation	TLS/SSL, JPEG, ASCII (Data)
5. Session	Establishes, maintains, and synchronizes dialog sessions	NetBIOS, RPC, PPTP (Data)

Lower Layers (Data Transport & Media Oriented)

The lower layers handle end-to-end transmission, logical routing, hardware framing, and physical signaling.

Layer	Primary Function	Protocols / Data Unit
4. Transport	End-to-end reliable delivery, flow and error control	TCP, UDP (Segment)
3. Network	Logical addressing, path determination, and routing	IPv4, IPv6, ICMP (Packet)
2. Data Link	Physical node-to-node framing and MAC access	Ethernet, Switches (Frame)
1. Physical	Transmits raw unstructured bit streams over media	Cables, Hubs, RJ45 (Bits)

Data Encapsulation Principle

During transmission, each layer wraps data received from the layer above with a protocol header (and a trailer at Layer 2), encapsulating the payload until it reaches the physical medium as electrical, optical, or radio bit signals.



TCP/IP PROTOCOL ARCHITECTURE

TCP/IP Four-Layer Model

The TCP/IP protocol suite is the pragmatic, open-standard architecture powering the modern Internet. Unlike the theoretical 7-layer OSI model, TCP/IP condenses networking functions into four practical layers.

TCP/IP Layer	Key Protocols and Operational Responsibilities
Application	HTTP, HTTPS, DNS, FTP, SMTP, SSH, DHCP — User communication and application logic.
Transport	TCP, UDP — End-to-end process-to-process communication, segmentation, and reliability.
Internet	IP (IPv4/IPv6), ICMP, ARP, IGMP — Logical packet addressing, path routing, and packet delivery.
Network Access	Ethernet (802.3), Wi-Fi (802.11), MAC, PPP — Hardware framing and physical transmission.

Comparison: OSI vs. TCP/IP Architecture

While the OSI model is widely referenced for educational and architectural analysis, TCP/IP represents the actual operational protocol stack implemented across modern commercial operating systems.

Comparison Parameter	OSI Reference Model	TCP/IP Architecture
Layer Count	7 distinct layers	4 functional layers
Protocol Coupling	Strict protocol independence	Protocols designed together
Session & Presentation	Separated into dedicated layers	Integrated into Application layer
Implementation	Theoretical conceptual framework	De facto global Internet standard

Why TCP/IP Dominates

- Exceptional interoperability across diverse vendor hardware architectures.
- Highly resilient dynamic packet routing that survives individual link failures.
- Universal global standard supported natively by all modern operating systems.



TRANSMISSION MEDIA

Guided Media (Wired)

Guided transmission media provide a physical conductor or waveguide to channel electromagnetic signals directly from the transmitter to the receiver with low interference and high security.

Cable Type	Bandwidth & Speed	Typical Engineering Application
Twisted Pair (UTP/STP)	10 Mbps – 10 Gbps (Cat5e/Cat6a)	Office LAN cabling, Ethernet switch interconnections.
Coaxial Cable	10 Mbps – 1 Gbps (DOCSIS)	Cable television distribution, broadband internet access.
Fiber Optic Cable	10 Gbps – 100+ Gbps	Telecommunication backbones, long-haul subsea lines.

Unguided Media (Wireless)

Unguided transmission media broadcast electromagnetic waves through the atmosphere or free space without physical conductors, providing mobility and flexible coverage.

Medium / Band	Frequency Range	Characteristics and Usage
Radio Waves	3 kHz – 1 GHz	Omnidirectional; penetrates walls; AM/FM and cellular bands.
Microwaves	1 GHz – 300 GHz	Unidirectional line-of-sight; satellite communication, radar.
Infrared	300 GHz – 400 THz	Short-range line-of-sight; remote controls, optocouplers.
Wi-Fi (RF)	2.4 GHz / 5 GHz / 6 GHz	High throughput local wireless data networking (802.11ax).

Example Comparison

Fiber Optic: Transmits light pulses via total internal reflection; completely immune to EMI with ultra-low attenuation over 40+ km.

Twisted Pair: Relies on copper wire twisting to cancel electromagnetic crosstalk; economical and easy to terminate, but limited to 100 meters.



DATA LINK LAYER & MAC SUBLAYER

What is the Data Link Layer?

The Data Link Layer is responsible for establishing reliable, point-to-point and point-to-multipoint data transfer across a physical link. It groups network layer packets into discrete frames, handles physical addressing via MAC addresses, and resolves shared channel access contention.

Common Functions of the Data Link Layer

Function	How It Operates	Mechanism / Protocol
Framing	Delineates packet data with start/end byte flags and headers	Ethernet II frame (preamble, SFD, FCS)
Error Detection	Detects bit corruption caused by electrical noise or attenuation	Cyclic Redundancy Check (CRC-32)
Flow Control	Regulates transmission rate to prevent receiver buffer overflow	Sliding Window, Pause frames (802.3x)
Media Access	Coordinates multiple nodes sharing a single broadcast channel	CSMA/CD (Ethernet) & CSMA/CA (Wi-Fi)

Steps in CSMA/CD Operation

- **Carrier Sense:** The transmitting station listens to the shared bus to determine if another host is currently transmitting.
- **Multiple Access:** If the channel is detected idle, the station begins transmitting its frame immediately.
- **Collision Detection:** While transmitting, the station continuously monitors channel voltage to detect overlapping signals.
- **Jam Signal & Backoff:** If a collision occurs, a jam signal is broadcast, and nodes execute a truncated binary exponential backoff.

Why MAC Addressing is Crucial

A 48-bit Media Access Control (MAC) address is permanently burned into the Network Interface Card (NIC) hardware, ensuring unambiguous local frame delivery across layer-2 switches regardless of higher-level IP configuration changes.



NETWORK LAYER & IP ADDRESSING

What is IP Addressing?

The Network Layer provides logical end-to-end addressing and packet routing across diverse interconnected subnets. An Internet Protocol (IP) address uniquely identifies each host network interface across the global internetwork.

IPv4 Classful Addressing Architecture

Class	First Octet Range	Default Subnet Mask	Target Network Scale
Class A	1 – 126	255.0.0.0 (/8)	Very large networks (up to 16M hosts)
Class B	128 – 191	255.255.0.0 (/16)	Medium to large corporate networks (65k hosts)
Class C	192 – 223	255.255.255.0 (/24)	Small local business/home LANs (254 hosts)
Class D	224 – 239	Reserved for Multicast	Multicast streaming and routing protocols
Class E	240 – 255	Reserved for Research	Experimental research and future allocation

Common Patterns in Modern IP Networking

- **Subnet Masking:** Splits host bits into subnet identifiers to partition broadcast domains.
- **CIDR (Classless Routing):** Uses prefix notation (e.g. /27) to prevent address pool exhaustion.
- **Private IP Spaces (RFC 1918):** Non-routable ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16).
- **IPv6 Transition:** 128-bit hexadecimal addressing providing 3.4×10^{38} globally unique IPs.

Benefit

Structured subnetting and hierarchical IP addressing dramatically shrink global router routing tables, isolate network broadcast traffic, and enforce fine-grained firewall security boundaries.



ROUTING PROTOCOLS

Routing Fundamentals

Routing is the mechanism by which routers inspect destination IP addresses, query dynamically maintained routing tables, and forward packets along the most optimal path across interconnected autonomous networks.

Comparison of Major Routing Protocols

Protocol	Algorithm Type	Routing Metric & Best-Fit Application
RIP (v1/v2)	Distance Vector (Bellman-Ford)	Hop count (max 15 hops); small, legacy campus networks.
OSPF	Link State (Dijkstra's SPF)	Path cost based on link bandwidth; large enterprise intranets.
EIGRP	Advanced Distance Vector (DUAL)	Composite metric (bandwidth, delay); Cisco enterprise environments.
BGP	Path Vector (Policy-based)	AS-Path and policy attributes; core routing across the global Internet.

Static vs. Dynamic Routing

Static routing requires network administrators to manually define forwarding paths in the routing table. Dynamic routing protocols automatically exchange link status updates, calculate optimal topologies, and immediately reroute traffic around severed communication cables or offline routers.

Using Routing Protocols Carefully

- Select protocols matching network scale, convergence requirements, and memory resources.
- Configure route summarization at boundary routers to prevent routing table bloat.
- Enforce cryptographic routing update authentication (e.g. MD5/SHA) to thwart route poisoning attacks.

TRANSPORT LAYER PROTOCOLS

What is a Transport Protocol?

The Transport Layer provides logical end-to-end communication between application processes running on distinct hosts. It accomplishes multiplexing and demultiplexing via port numbers, manages packet segmentation, and enforces delivery contracts.

Detailed Comparison: TCP vs. UDP

Parameter	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (3-way handshake)	Connectionless (no setup handshake)
Reliability	Guaranteed delivery (ACKs & retransmission)	Best-effort; packets may drop or duplicate
Ordering	Sequenced packet reassembly at destination	No sequencing; arrives in any order
Header Size	20 – 60 bytes (flags, window, options)	8 bytes fixed (minimal overhead)
Flow Control	Dynamic sliding window and congestion control	No flow control or congestion mitigation
Applications	Web (HTTP), Email (SMTP), File transfer (FTP)	DNS, VoIP, Live video streaming, Gaming

Why Learn Both Protocols?

- TCP provides absolute reliability and sequencing for critical data exchanges.
- UDP delivers ultra-low latency essential for real-time video, audio, and gaming.
- Helps software engineers select optimal transport models based on application tolerance.



APPLICATION LAYER SERVICES

Academic & Core Services

The Application Layer directly interfaces with user-facing software applications, providing essential network services, session negotiations, domain translations, and standardized message structures.

Protocol	Standard Port	Primary Operational Function
HTTP / HTTPS	80 / 443	Transfers web pages and REST API payloads with SSL/TLS encryption.
DNS	53	Translates human-readable domain names into 32-bit/128-bit IP addresses.
DHCP	67, 68	Automates dynamic client IP address assignment and gateway discovery.
FTP / SFTP	20, 21 / 22	Transfers bulk files between clients and remote storage servers.
SMTP / IMAP	25 / 143	Routes outgoing mail between MTAs (SMTP) and retrieves user mail (IMAP).

Enterprise Architecture Models

Network applications are primarily structured around client-server hierarchies or peer-to-peer distributed topologies.

Architecture Model	Structural Characteristics and Real-World Example
Client-Server	Centralized service provider listens for client connections; highly scalable (e.g. Apache Web Server, AWS Cloud).
Peer-to-Peer (P2P)	Decentralized nodes act simultaneously as client and server; resilient to single-point failure (e.g. BitTorrent, IPFS).

Good Practice

- Enforce encrypted HTTPS/TLS across all public and internal service communications.
- Implement TTL caching for DNS queries to reduce latency and recursive resolver load.



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NETWORK SECURITY & HARDWARE DEVICES

Network Security Fundamentals

- Deploy Next-Generation Firewalls (NGFW) to inspect stateful connections and deep packet contents.
- Implement IPsec and SSL/TLS Virtual Private Networks (VPNs) for secure remote access tunnels.
- Segment internal network domains using Virtual LANs (VLANs) and access control lists (ACLs).
- Enforce robust symmetric and asymmetric encryption (AES-256, RSA) across data at rest and in transit.
- Utilize Intrusion Detection and Prevention Systems (IDS/IPS) to detect anomalous traffic patterns.
- Maintain continuous security patch management across all active switch and router firmware.

Core Networking Devices and Functions

Device	Operating Layer	Core Architectural Function
Repeater / Hub	Layer 1 (Physical)	Regenerates bit signals; broadcasts incoming data blindly to all ports.
Bridge / Switch	Layer 2 (Data Link)	Learns MAC addresses dynamically; forwards frames selectively without collisions.
Router	Layer 3 (Network)	Connects disparate subnets; routes IP packets based on routing table algorithms.
Firewall	Layers 3, 4, 7	Filters unauthorized ingress and egress traffic based on predefined security rules.
Gateway	Layers 4 – 7	Translates protocols between fundamentally incompatible network architectures.